AN ALGEBRAIC SPACE OVER A FIELD IS UNIVERSALLY OPEN OVER THAT FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof uses the standard device of reducing universal openness to a surjective etale presentation plus the local criterion; both steps are written out.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- A scheme over a field is universally open over that field. Morphisms, Lemma morphisms-lemma-scheme-over-field-universally-open, used at proof lines 1084-1087. Gives openness of the composite U -> Spec(k).
- Surjectivity is stable under base change. Lemma lemma-base-change-surjective, used at proof lines 1088-1091. Makes U x_k Z -> X x_k Z surjective.
- The local criterion for universal openness of a morphism of algebraic spaces. Lemma lemma-universally-open-local, used at proof lines 1096-1098. Concludes universal openness of p.
- Representing a morphism of algebraic spaces by schemes through an etale cover. used at proof lines 1083-1084. Supplies the scheme U mapping surjectively etale onto X.

Source reference. Original lemma 03MH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4749–4749. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03Z5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5208–5208. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0383: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4222–4222. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $p : X \to \mathrm{Spec}(k)$ be a morphism of algebraic spaces over S where k is a field. Then $p : X \to \mathrm{Spec}(k)$ is universally open.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Choose a scheme U and a surjective étale morphism $U \to X$. The composition $U \to \mathrm{Spec}(k)$ is universally open (as a morphism of schemes) by Morphisms, Lemma 0383. Let $Z \to \mathrm{Spec}(k)$ be a morphism of schemes. Then $U \times_{\mathrm{Spec}(k)} Z \to X \times_{\mathrm{Spec}(k)} Z$ is surjective, see Lemma 03MH. Hence the first of the maps

$$|U \times_{\mathrm{Spec}(k)} Z| \to |X \times_{\mathrm{Spec}(k)} Z| \to |Z|$$

is surjective. Since the composition is open by the above we conclude that the second map is open as well. Whence p is universally open by Lemma 03Z5. $\square$